

Class Meeting Handout #19

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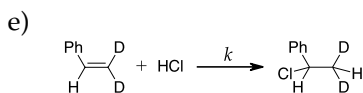
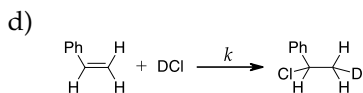
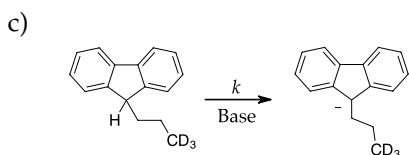
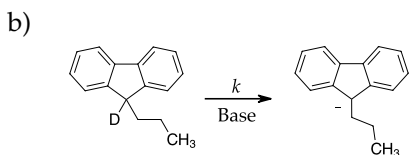
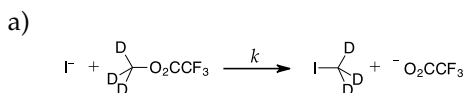
We have been introduced to the theory of kinetic isotope effects. Today we will explore some examples of mechanistic interpretation using this important tool of physical organic chemistry.¹

Kinetic Isotope Effects

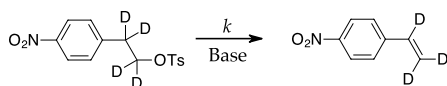
Let us practice our skills. You will want to review our discussion from the previous class meeting.²

Exercise: Using the Kinetic Isotope Effect – 15 minutes

1. Predict the type and magnitude of the kinetic isotope effects for the reactions listed in table 1 with substitutions of the indicated protons with deuterium? Write a brief defence of each decision.



2. For the following reaction, propose structures for the transition state in the rate-determining step that would be consistent with the isotope effects listed in table 2. Indicate which bonds are breaking or forming in the T.S. of each reaction. What type of reaction describes each mechanism (i.e. S_N1 , etc.). Which of the three set of data below is likely observed in this reaction and which two sets are likely fictional?



This document was produced using the $\text{\LaTeX}$ typesetting language with the Tufte-handout document class. Chemical diagrams were created in *ChemDoodle* Plots were typeset using *Matplotlib* and *Python* tools. Diagrams were created and edited using *Affinity Designer*.

¹ Read chapter 8.1.

² Read chapter 8.1 of the textbook and read the Kinetic Isotope Effect tutorial document available on the Moodle page.

Table 1: Cases for problem #1.

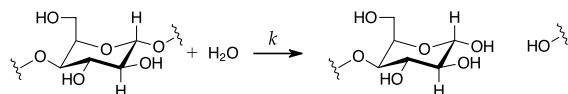
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Table 2: Cases for problem #2. ↓

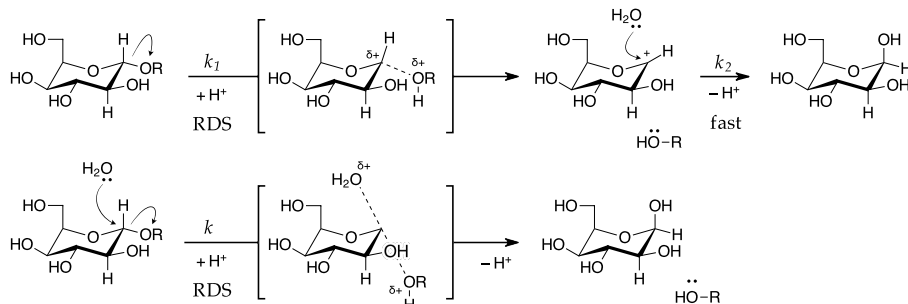
- a) $k_H/k_D = 5.2$
 $k_{O_{16}}/k_{O_{18}} = 1.013$
- b) $k_H/k_D = 1.3$
 $k_{O_{16}}/k_{O_{18}} = 1.016$
- c) $k_H/k_D = 4.3$
 $k_{O_{16}}/k_{O_{18}} = 1.000$

Exercise: Interpreting an Enzyme Reaction Mechanism – 15 minutes

3. For the glycosidic cleavage of the bacterial cell wall catalyzed by *lysozyme*, the following reaction occurs...



It is proposed that the reaction proceeds by one of two mechanisms...



- Describe each mechanism (or each step in each mechanism) using the common designations (i.e. S_N1 , etc.).
- Design an experiment that involves isotopic substitution of hydrogen atoms that will elucidate the mechanism. Describe the isotopic substitutions you would make.
- Describe the expected results for your experiment for each mechanism.
- In addition to your experiments above, explain what the expected primary kinetic isotope effect would be for isotopic substitution of the oxygen in water in each scheme.
- Explain what the expected primary kinetic isotope effect would be for isotopic substitution of the oxygen in the hydroxyl leaving group in each scheme.

Discussion: pH in D₂O – 10 minutes

- How might an enzyme reaction such as the above be affected by performing it in D₂O instead of water?
- Enzyme reactions are usually pH-sensitive. What is the effect of D₂O on acid equilibrium?
- If your pH-meter reads 7.00 when in a solution of D₂O, is the solution neutral? How do we relate pD to pH when measured by an electrode?

Learning Goals

Through the suggested textbook reading, problems and participating in this class discussion, students will...

- be able to express the definitions involved in isotope effects
- be able to identify primary and secondary isotope effects by examining a proposed mechanism.
- be able to predict the magnitude of an isotope effect in an experimental design and interpret experimental results.
- be able to accurately measure the protonating power (pD) in D_2O using a pH electrode.

A Look Ahead

We have completed the first part of the course. Congratulations! These first four lessons reminded you of your existing skills in chemical mechanism and introduced you to new methods and tools of physical organic chemistry. In the second part of the course we will examine examples of mechanism in organic chemistry and how physical organic chemistry has helped to elucidate these mechanisms.

We will be relying on the examples in the textbook and using literature examples. We will explore how to critically evaluate published data. Soon you will be choosing a topic for your own detailed exploration of a body of work in the recent literature.³

But, before we get to any of that, you must endure your first test.

³ Remember that the definition of “recent” in physical organic chemistry pretty much starts with the day your grandfather was born.